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Research paper

Numerical simulation of a GM-type pulse tube cryocooler system: Part II. Rotary valve and cold head

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ABSTRACT

A simplified model to characterise a rotary valve is proposed. This model is used inside a complete GM-type pulse tube cryocooler (GMPTC) simulation including the compressor, rotary valve and cold head. The verification of the model is presented including the pressure wave, cooling power and COP by comparing the simulation results to the experimental data of an existing GMPTC system. The exergy losses produced by the compressor, rotary valve and cold head are discussed. This paper is aimed to propose a simulation method which can be applied to analyse and optimise a GMPTC cold head by using the characteristics of a specific compressor and rotary valve.

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1. Introduction

As GM-type pulse tube cryocoolers (GMPTCs) become competitive to GM cryocoolers in various applications, a closer look at their overall efficiency could be worthwhile. To predict the performance of GMPTCs, many theoretical analysis methods, as well as numerical models have been developed. Most of these works were focused on the cold head without considering the effect of the compressors and switching valves. Often the boundary pressure wave (usually at the warm end of the regenerator) was preset as a fixed square [1], sinusoidal [2] or other type [3,4] wave function.

However, in a real system, the pressure wave is a dynamic variable with irregular form [5,6] and will change with the working condition, e.g. cooling temperature, filling pressure, or valve setting. The pressure wave is also heavily influenced by the characteristics of the compressor and the switching valve. So it makes sense to simulate the complete system of a GMPTC to be able to visualise the exergy loss distribution in order to improve the performance.

Zhu et al. [7] developed a numerical model including a switching valve to analyse the working process of an active-buffer GMPTC with fixed discharge and suction pressures of the compressor. Pfotenhauer and Baik [8] characterised a compressor for designing a GMPTC. In this work, the GMPTC regenerator was optimised with the REGEN 2.0 simulation software [9], which treats the pressure wave and mass flow as sinusoidal functions. Furthermore, the phenomena that mass-distribution of the working gas inside a GMPTC changes with the cooling temperature T_c (as discussed in Part I [10]) was ignored in these works. Therefore it is thought to be necessary to build a complete GMPTC model which could reproduce the behaviour of a real system more closely.

In the present work, a method to characterise a rotary valve is proposed. Then the model of a complete GMPTC system is built, taking the characteristics of compressor [10] and rotary valve into account. The model is based on the commercial software Sage [11] and verified by comparing the pressure wave, cooling power $\dot{Q}_c$ and COP of the simulation results to the experimental data of an existing GMPTC system. Finally, the exergy losses induced by the compressor, rotary valve, and cold head are discussed.

2. Model of the rotary valve

A rotary valve is commonly used as a switching valve to transform the steady working gas flow (DC flow) produced by the compressor to an oscillating flow (AC flow) usable for the cold head. The examined rotary valve was developed at University of Giessen and produced by TransMIT GmbH; it is similar to the one shown in Fig. 1 of Ref. [12]. There are two circular passages on the stator, which are alternately linked to the discharge and the suction pres-

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Nomenclature

T	temperature (K)
p	pressure (Pa)
ρ	density (kg/m^3)
h	specific enthalpy (J/kg)
s	specific entropy (J/kg K)
V	volume (m^3)
$\dot{W}$	power (W)
$\dot{X}$	rate of exergy (W)
Δp	pressure difference (Pa)
$\dot{m}$	mass flow (kg/s)
m	mass (kg)
R	gas constant (J/mol K)
n	molar quantity (mol)
M	molar mass (kg/mol)
$\dot{V}$	volume flow (m^3/s)
P_r	pressure ratio (–)
t	time (s)
r	time ratio of a cycle (–)
F	flow area multiplier (–)
f	frequency (Hz)
R_{el}	exergy loss ratio (–)
R_{ac}	ratio of the acoustic power required for cooling to the electrical input power
C_d	discharge coefficient (–)
$\dot{Q}_c$	cooling power (W)

Greek

η_{ex}	exergetic efficiency of the compressor
ω	angular velocity (degree/s)
α	phase angle for the period that the open ratio starts from zero to one in one cycle (degree)

Subscripts

d	discharge condition
s	suction condition
0	room condition
it	isothermal process
e	electrical
i	ideal gas
f	filling condition
db	discharge buffer
sb	suction buffer
dl	discharge flexible line
sl	suction flexible line
tot	total amount in the system
com	compressor package
rotary	rotary valve
c	cold tip of the cryocooler
ac	acoustic
m	mean value
min	minimum value
reg	regenerator
pt	pulse tube
Carnot	Carnot cycle

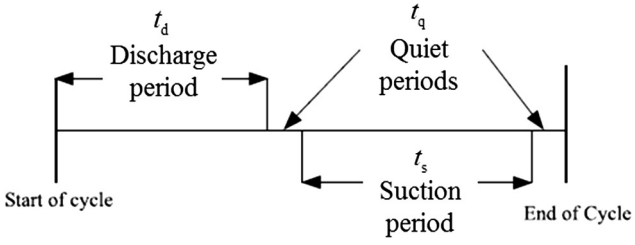


Fig. 1. The operation cycle controlled by the rotary valve [13].

sure sides of the compressor package. Each of the circular passages has the diameter of 5.2 mm. The alternating process is realised by a rotor. Usually the operation cycle will be divided into 4 parts, a discharge, a suction and two quiet periods, as shown in Fig. 1. The time ratio is taken as [13]

$$r = \frac{t_s}{t_d} = 1.231, \tag{1}$$

In the Sage model, the rotary valve can be represented by two time-dependent valves noted as "discharge valve" and "suction valve" in Fig. 2. Both valves are linked to the cold head on the right side (not shown). The diameter of each valve is set as the hydraulic diameter of the two circular passages in the real rotary valve. A detailed description of the compressor model is given in Part I [10].

In Sage, the time ratio can be adjusted by changing the flow area multipliers F_d and F_s of the "discharge valve" and "suction valve", respectively. These multipliers are in a typical range between 1 (fully open) and 0 (fully closed). In order to avoid too many harmonics, the flow area multipliers are modelled as sine wave functions with the time ratio r and phase shift α as parameters:

$$F_d = \begin{cases} F_d(\omega t) & \text{when } F_d(\omega t) > F_{min} \\ F_{min} & \text{when } F_d(\omega t) \leqslant F_{min} \end{cases} \tag{2}$$

$$F_s = \begin{cases} F_s(\omega t) & \text{when } F_s(\omega t) > F_{min} \\ F_{min} & \text{when } F_s(\omega t) \leqslant F_{min} \end{cases} \tag{3}$$

where

$$F_d(\omega t) = \frac{\cos(\omega t - \alpha) - \cos\alpha}{1 - \cos\alpha} \tag{4}$$

$$F_s(\omega t) = \frac{\cos(\omega t - \alpha) + \cos(r\alpha)}{\cos(r\alpha) - \cos[(r-1)\alpha]} \tag{5}$$

in which α is the time (in degrees, see Fig. 3) in which the flow area multiplier changes from zero to one. F_{min} is added as an additional parameter, to be able to limit the minimum flow area. It is a small number to account for the gap between rotor and stator in a real rotary valve. The functions F_d and F_s are plotted in Fig. 3 for F_{min} = 0.001. There is a short time in which the F_s will be larger than 1. Since deviation of the F_s is very small compared to 1 as well as the short period compared to the whole cycle, it is supposed to be negligible.

Up to now, the leakage factor F_{min} and discharge coefficient C_d (defined in the Chapter 32 of "Sage User's Guide" [11], inversely proportional to flow resistance) of the rotary valve are still undetermined. These parameters are important but difficult to include in the theory due to the unknown flow resistance of the rotary valve. Such measurements are planned for the future. Until then, these parameters are determined by comparing the simulated and measured pressure wave form as shown in the next section.

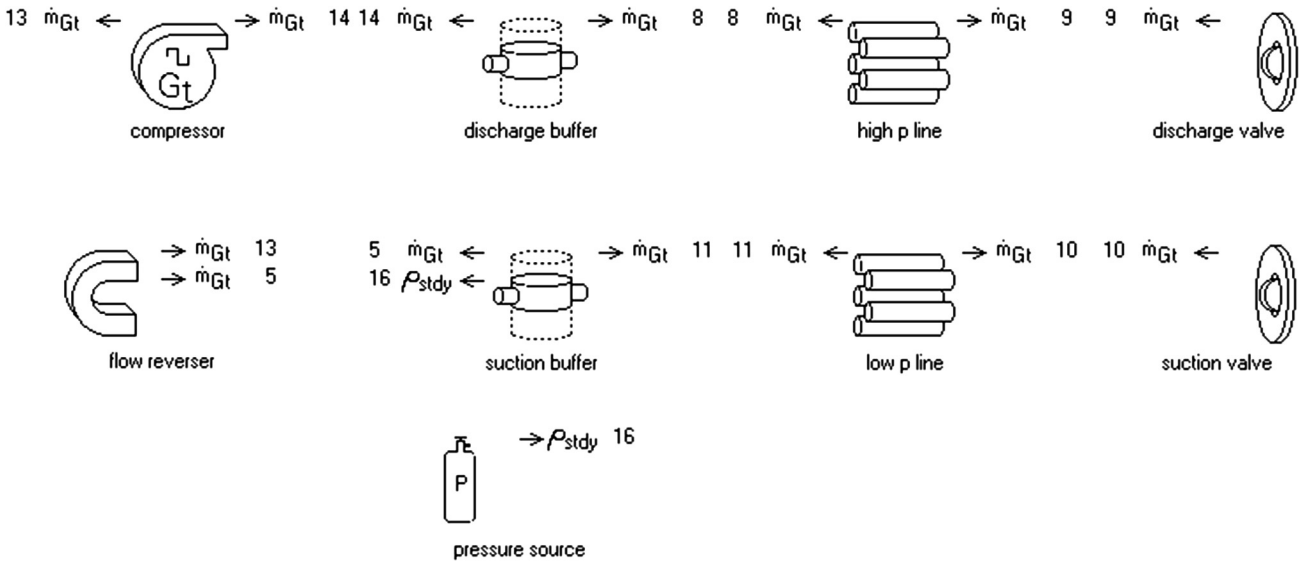


Fig. 2. Compressor and rotary valve model in Sage.

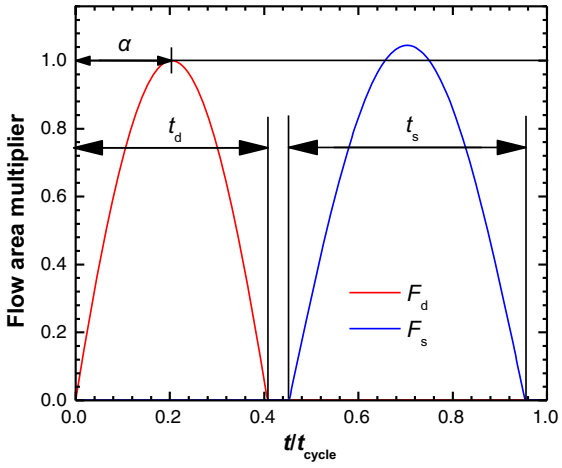


Fig. 3. The flow area multipliers for the discharge and suction valve model in one cycle.

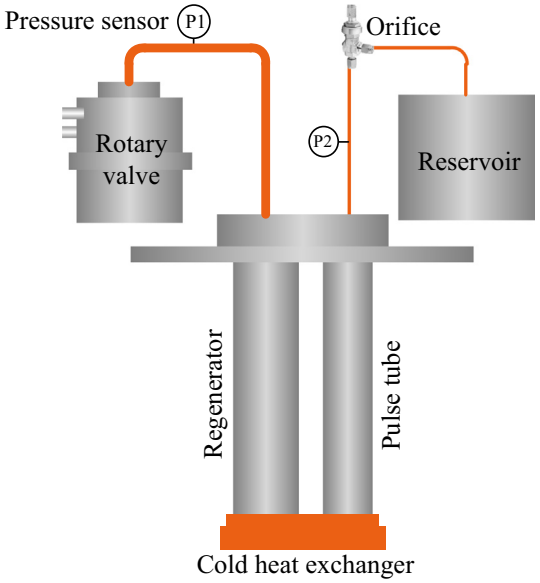


Fig. 4. Schematic diagram of the cold head and rotary valve.

3. Experiment and simulation of a GMPTC

The performance of an existing orifice type GMPTC cold head [14], as shown in Fig. 4, operated with a Leybold RW2 compressor was measured and simulated. The regenerator was filled with stainless steel mesh #247 only. The cold head was modelled with Sage based on the published dimensions of the cold head [14]. The filling pressure of the system was 1.7 MPa. In the experiment, the orifice (see Fig. 4) was adjusted for achieving the lowest no-load cooling temperature. Its discharge coefficient was set corresponding to its real opening, while the diameter was fixed at 1.42 mm in Sage according to the product manual of the real orifice valve (Swagelok SS-3MA-MM). During the simulation, the total mass of the helium working gas inside the system was fixed as a constraint to the model (see Eq. (1) in Part I [10]). Therefore the mean pressure in the cold head could be self-adjusted to satisfy that constraint for different working conditions during the simulation. For matching the pressure wave monitored by the pressure sensor P1 after the rotary valve as shown in Fig. 4, the leakage factor F_{min} and discharge coefficients C_d of the discharge and suction valves in the Sage model were set as 0.031 and 0.6 respectively. The simulated pressure waves, cooling power $\dot{Q}_c$ and COP results are compared with the experiment for verification in the next section.

3.1. Comparison between experiment and simulation

3.1.1. Pressure wave

As the pressure wave is the only internal parameter of the GMPTC system that can be easily measured for a whole cycle when the system is running, its simulated and experimental results are compared for different cooling temperatures T_c and frequencies f.

The pressure waves from the experiment and the Sage model before the regenerator (P1) and after the pulse tube (P2, see Fig. 4) are plotted in Figs. 5 and 6 for an operating frequency of 2.3 Hz and a no-load cooling temperature of 32.9 K, when the discharge coefficient of the orifice is 0.22. The simulated pressure waves are found to match the experimental data well except for

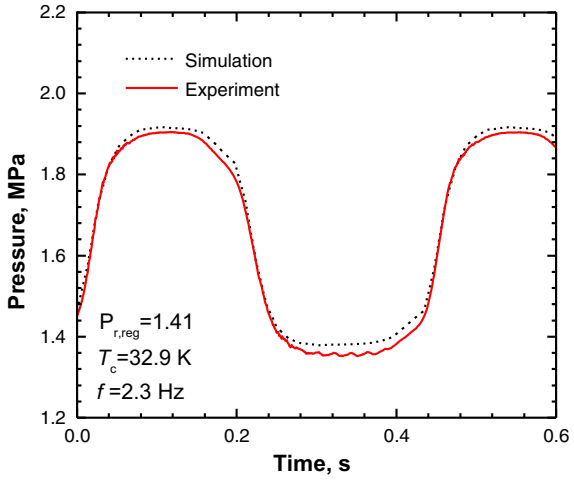


Fig. 5. Pressure wave after the rotary valve (f = 2.3 Hz, T_c = 32.9 K).

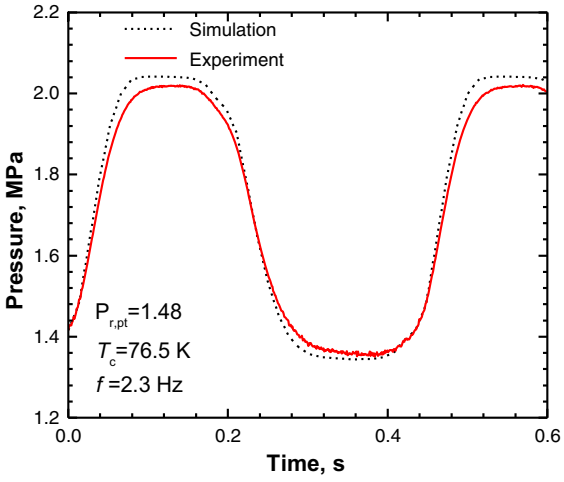


Fig. 8. Pressure wave at the pulse tube warm end (f = 2.3 Hz, T_c = 76.5 K).

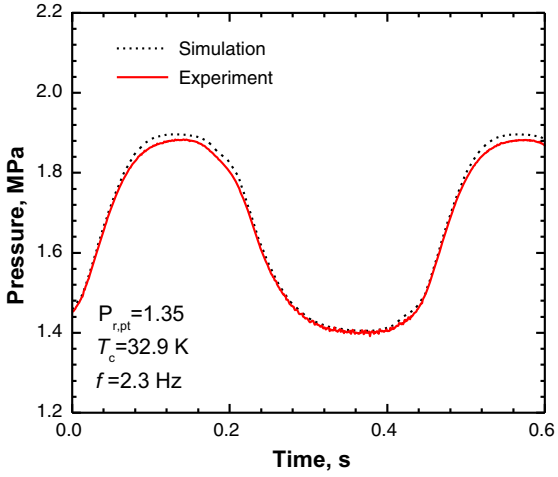


Fig. 6. Pressure wave at the pulse tube warm end (f = 2.3 Hz, T_c = 32.9 K).

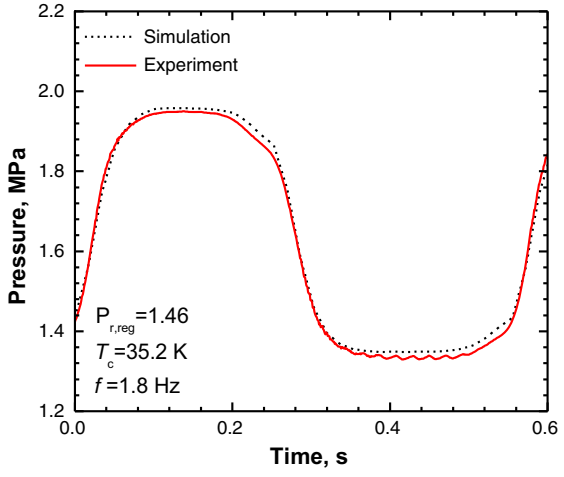


Fig. 9. Pressure wave after the rotary valve (f = 1.8 Hz, T_c = 35.2 K).

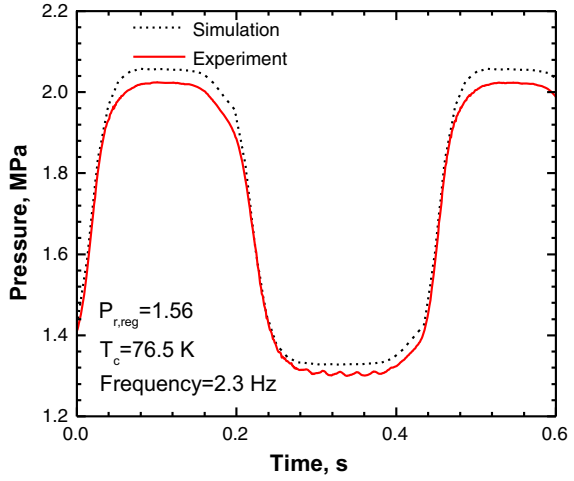


Fig. 7. Pressure wave after the rotary valve (f = 2.3 Hz, T_c = 76.5 K).

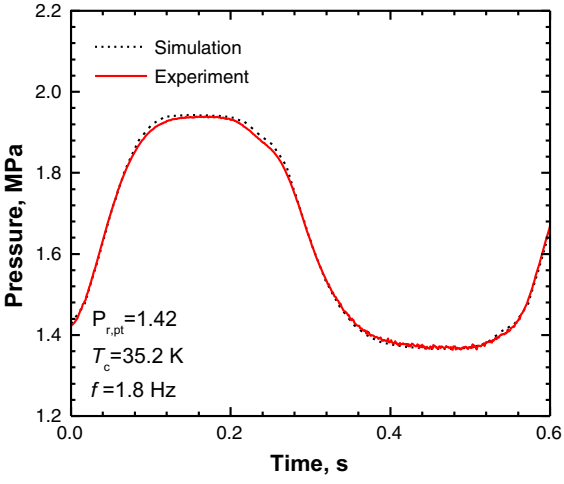


Fig. 10. Pressure wave at the pulse tube warm end (f = 1.8 Hz, T_c = 35.2 K).

the slight deviation of the mean pressure. When the cold head is warmed up to 76.5 K using an electrical heater at unchanged frequency and orifice setting, the pressure ratio becomes larger as shown in Figs. 7 and 8. That is because the flow resistance of the cold head increases, mainly caused by the increased viscosity of helium gas at higher temperatures. The measured pressure ratios are presented as $P_{r,reg}$ and $P_{r,pt}$ for P1 and P2 respectively on the figures and calculated as the maximum pressure divided by the minimum pressure. The mean pressure becomes higher mainly due to the lower density of the gas in the cold head.

Figs. 9 and 10 show the pressure waves with an operating frequency f of 1.8 Hz. The lowest no-load cooling temperature achieved is 35.2 K when the orifice valve is optimised, corresponding to a discharge coefficient of 0.175 in the model. The pressure ratio becomes larger compared to Figs. 5 and 6 with similar cold head temperatures. This is due to the increasing of the flow resistance, caused by the lower discharge coefficient of the orifice.

These comparisons show that the simulated pressure waves can fit the experimental data well with the preset leakage factor F_{min} and discharge coefficient C_d of the rotary valve. The form and amplitude of the simulated pressure waves under different cooling temperatures T_c and frequencies f reflect the real working conditions of the GMPTC system under test.

3.1.2. Cooling power and COP

In this section, the calculated cooling power $\dot{Q}_c$ and coefficient of performance (COP) from the Sage model are compared with the measurements, as shown in Figs. 11 and 12. Here the COP is defined as the cooling power $\dot{Q}_c$ divided by the electrical input power $\dot{W}_e$. Although the simulation results seem a little worse than the experiment, they are very close. Other than the environmental factors, this deviation can be induced mainly by the following ones:

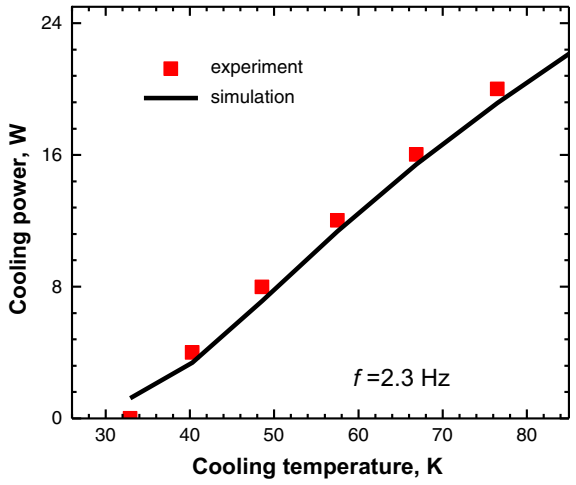


Fig. 11. Comparison of the cooling power $\dot{Q}_c$ of experimental and simulation data at f = 2.3 Hz.

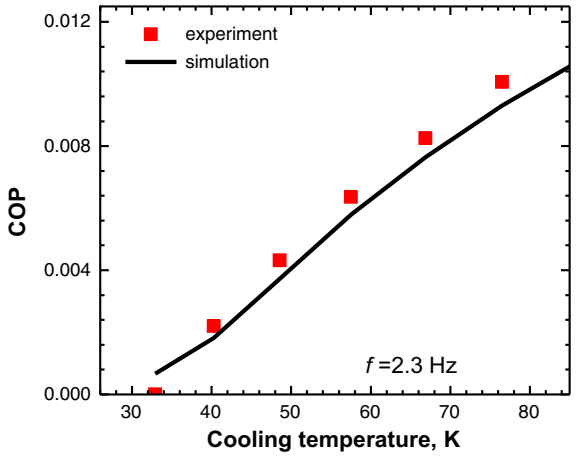


Fig. 12. Comparison of the COP of experimental and simulation data at f = 2.3 Hz.

1. The compressor model was built based on the steady flow condition measurement. When the compressor is used in the GMPTC system, there will be pressure fluctuations with small amplitudes both at inlet and outlet of the compressor. This could cause some error in the efficiency and the volume flow prediction, because we based our calculations on the mean values of suction and discharge pressures.
2. Our static compressor measurement did not take into account the various compliances and resistances inside the compressor package. In a dynamic operation with a rotary valve and a cryocooler attached, these compliances may affect the pressure wave and volume flows.
3. Leakage factor F_{min} and discharge coefficients C_d of the rotary valve were determined by fitting the pressure wave. Although the shape of the pressure waves from the Sage model and experiment are almost the same, the volume flow and the phase shift between volume flow and pressure wave are unknown.
4. The discharge coefficient of the orifice (needle valve) was set according to the manual provided by Swagelok for static flow [15]. The non-linear flow resistance of the valve cannot be matched exactly to the Sage representation. Especially under dynamic flow conditions as in the cryocooler system, the behaviour of the valve will be somehow different.
5. The streaming and shuttle losses in the pulse tube are ignored. In Sage, the equations to predict these losses are derived under the sine-wave pressure and mass flow assumptions [11]. Even in a Stirling-type pulse-tube cryocooler which meets the assumptions, these two losses calculated by a Sage model have been found to be different from those by CFD analysis [16]. However, for the GMPTC the pressure (see Figs. 5–10) and mass flow (Figs. 15 and 16) waves obviously deviate from a sine wave. So the applicability of streaming and shuttle-loss equations built in Sage is doubtful. Leaving them totally out gave the best conformance with the experiment in our case, but that may not hold for other cases. Switching these losses on will dramatically degrade the cooling performance in this GMPTC Sage model which is still inexplicable. For example, when these losses are switched on in the model, the cooling power at 76.5 K will be decreased to 10.8 W, which is only 54% of that (20 W, see Fig. 11) without these losses. Thus, further effort is needed to clarify these losses in the future.

However, based on the fine reproduction of pressure wave, cooling power and COP, the model is still supposed to give usable results. The exergy losses caused by the compressor, rotary valve and the cold head can be calculated and compared.

3.2. Exergy loss distribution

In this section, the exergy loss of each component will be evaluated as ratio of the lost exergy to the electrical input power $\dot{W}_e$. Then the exergy loss ratio in the compressor package can be expressed as one minus the exergetic efficiency [10] of the compressor

$$R_{el,com} = (\dot{W}_e - \dot{X}_{com})/\dot{W}_e = 1 - \eta_{ex}. \quad (6)$$

When the exergy loss through the transformation process in the rotary valve is evaluated, the basic definition [17] of acoustic power

$$\dot{W}_{ac} = f \cdot \oint (p - p_m)\dot{V}dt \quad (7)$$

is applied to calculate the exergy after the rotary by use of the discrete results of pressure p and volume flow $\dot{V}$ from the Sage model.

The exergy loss of the rotary valve can be treated as the difference between the exergy provided by the compressor $\dot{X}_{com}$ and the acoustic power after the rotary valve $\dot{W}_{ac,rotary}$. Then the exergy loss ratio of the rotary valve is given by

$$R_{el,rotary} = (\dot{X}_{com} - \dot{W}_{ac,rotary})/\dot{W}_e. \quad (8)$$

For an ideal cold head with Carnot COP, The acoustic power required to produce a certain cooling power can be written as

$$\dot{W}_{ac,Carnot} = \dot{Q}_c \frac{T_0 - T_c}{T_c}, \quad (9)$$

where T_0 is the room temperature. Then the exergy loss ratio of the cold head can be expressed as

$$R_{el,c} = (\dot{W}_{ac,rotary} - \dot{W}_{ac,Carnot})/\dot{W}_e. \quad (10)$$

When the ratio of the acoustic power required for cooling (with Carnot COP) to the electrical input power is defined as

$$R_{ac,Carnot} = \dot{W}_{ac,Carnot}/\dot{W}_e, \quad (11)$$

combining Eqs. (6), (8), (10) and (11), the sum of the ratios equals 1, namely

$$R_{el,com} + R_{el,rotary} + R_{el,c} + R_{ac,Carnot} = 1. \quad (12)$$

With these definitions, the exergy losses of the compressor, rotary valve and cold head can be determined quantitatively. For example, as shown in Fig. 13, when the cooling temperature T_c is 76.5 K and the operation frequency f is 2.3 Hz, the loss in the cold head is only 8.92% while those of the compressor and rotary valve are 65.17% and 23.20% respectively. So the performance of the complete GMPTC system is mainly determined by the efficiency of the compressor and the rotary valve.

The highest efficiency of the compressor RW2 used in our experiments is about 35% according to Fig. 4 in Part I [10]. The only way to lower this part of the exergy loss is using a more efficient compressor. However every compressor model achieves its highest efficiency at different pressure ratios (see Fig. 5 in Part I [10]). Therefore, the cold head needs to be optimised to work with a pressure ratio, where the efficiency of the compressor is near its maximum value.

For the rotary valve, the friction and the leakage are supposed to be the main factors to cause the exergy loss. In the Sage model, the leakage can be limited to a very small value by setting F_{min} of the rotary valve close to zero like 0.001. Then the exergy loss distribution can be re-plotted as displayed in Fig. 14. The loss of the rotary valve becomes much smaller compared to Fig. 13. The exergy loss in the compressor rises a bit because the pressure ratio becomes larger as the leakage goes down. Finally, the required acoustic power for cooling is increased by about 35% from 2.71% (Fig. 13) to 3.66% (Fig. 14), and thereby also the COP of the complete GMPTC

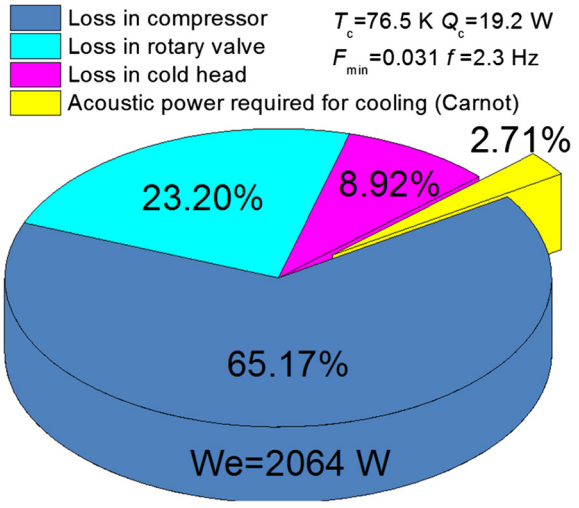


Fig. 13. Exergy loss distribution for F_{min} = 0.031.

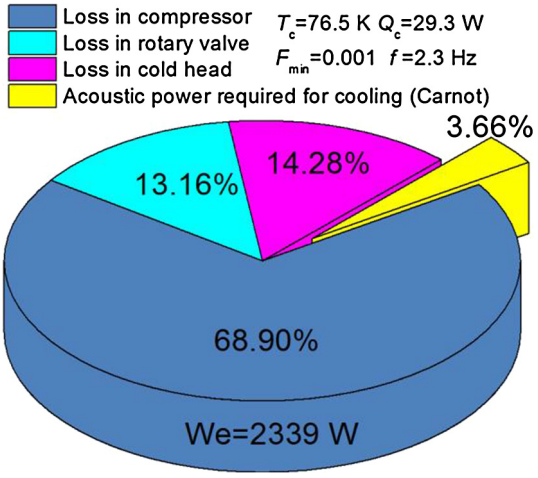


Fig. 14. Exergy loss distribution for F_{min} = 0.001.

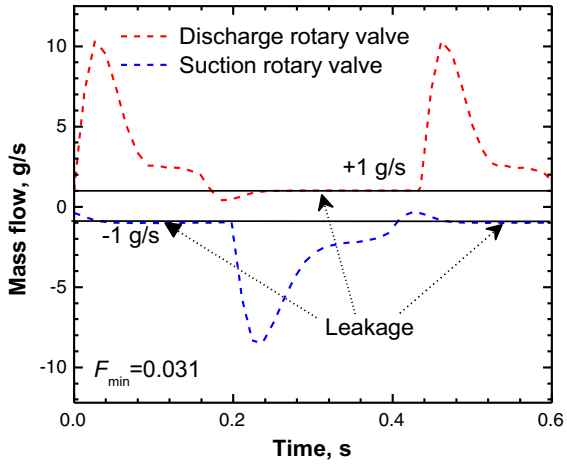


Fig. 15. Discharge and suction mass flow for F_{min} = 0.031.

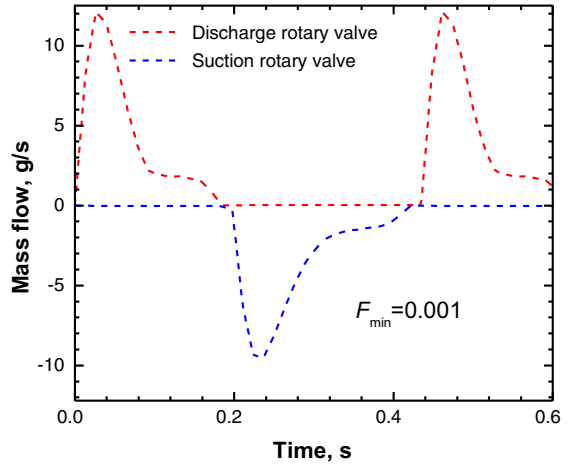


Fig. 16. Discharge and suction mass flow for F_{min} = 0.001.

system is increased by the same amount. The mass flow through the suction valve and discharge valve are plotted in Figs. 15 and 16 corresponding to the cases shown Figs. 13 and 14, respectively. A positive value means the direction of the mass flow is from the compressor to cold head while a negative one is from cold head to compressor. When $F_{min} = 0.031$, the leakage can be as high as 1 g/s as seen from Fig. 15. This mass flow leakage through the rotary valve is about 43% of the total mass flow produced by the compressor. This rather high leakage, which was determined by fitting the simulation results to the experiment, needs to be verified by further measurements. It is also possible that this leakage factor arises from errors in the compressor or cold head model, or a problem with the real rotary valve used in the tests. When $F_{min} = 0.001$, the leakage of rotary valve becomes almost zero as shown in Fig. 16 and now the higher peak values of the mass flow induce more friction loss. There is still 13.16% exergy lost in the rotary valve which is comparable to the loss in the cold head (Fig. 14). Because this friction loss cannot be avoided with this kind of rotary valve, it is important to make the leakage of a real rotary valve as low as possible. In addition, optimizing of the diameters or the shapes of the passages in the rotary valve to achieve a lower friction loss should be seriously considered.

4. Summary

A numerical simulation model of a complete GM-type pulse tube cryocooler system based on the Sage software was developed. This model was able to reproduce the measured cooling power and COP with good accuracy.

Based on the simulation results, a total system exergy loss analysis was performed, which shows that the main losses come from the inefficient GM compressor. The second largest loss arises from the rotary valve, which can be divided into two loss mechanisms: friction and leakage. Our simulations show a rather high leakage loss, whose origin is unknown. A better model of the rotary valve needs to be developed by further experiments in the future in order to avoid the indirect determination of the rotary valve's characteristics.

This work shows that it is important to measure the characteristics of the compressor and the rotary valve before designing a GM pulse tube cryocooler. The simulation method proposed by this paper can be a useful tool to optimise a cold head.

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References

[1] Xu MY, He YL, Chen ZQ. Analysis of an orifice pulse tube refrigerator using the method of characteristics. Cryogenics 1999;39(9):751–7.
[2] Banjare YP, Sahoo RK, Sarangi SK. CFD simulation and experimental validation of a GM type double inlet pulse tube refrigerator. Cryogenics 2010;50(4):271–80.
[3] Yang LW, Zhou Y, Liang JT. DC flow analysis and second orifice version pulse tube refrigerator. Cryogenics 1999;39(3):187–92.
[4] Wang C. Numerical analysis of 4 K pulse tube coolers: Part I. Numerical simulation. Cryogenics 1997;37(4):207–13.
[5] Gan ZH, Dong WQ, Qiu LM, Zhang XB, Sun H, He YL, et al. A single-stage GM-type pulse tube cryocooler operating at 10.6 K. Cryogenics 2009;49(5):198–201.
[6] Wang C, Thummes G, Heiden C. Control of DC gas flow in a single-stage double-inlet pulse tube cooler. Cryogenics 1998;38(8):843–7.
[7] Zhu SW, Kakimi Y, Matsubara Y. Investigation of active-buffer pulse tube refrigerator. Cryogenics 1997;37(8):461–71.
[8] Pfotenhauer JM, Baik JH. Compressor-specific design of a single stage pulse tube refrigerator. In: Cryocoolers 11. Keystone (Colorado): Kluwer Academic Publishers; 2002. p. 249–57.
[9] Arp V, Radebaugh R. Interactive program for microcomputers to calculate the optimum regenerator geometry for cryocoolers (REGEN and REGEN2). AFWAL Rep. AFWAL-TR-87-3040 ed.; 1987.
[10] Liu DL, Dietrich M, Thummes G, Gan ZH. Numerical simulation of a GM-type pulse tube cryocooler system: Part I. Characterization of compressors. Cryogenics 2017;81:8–13.
[11] Gedeon D. Sage User's Guide. Sage v10 ed.; 2014.
[12] Will ME, Tanaeva IA, Li R, de Waele ATAM. New rotary valves for pulse-tube refrigerators. Cryogenics 2004;44(11):793–800.
[13] Qiu LM, Thummes G. Valve timing effect on the cooling performance of a 4 K pulse tube cooler. Cryogenics 2002;42(5):327–33.
[14] Jiang YL, Chen GB, Thummes G. Experimental investigation on DC flow suppression and refrigeration characteristics of a high-performance single-stage GM-type PTC. In: Proceedings of the 20th International Cryogenic Engineering Conference (ICEC 20). Elsevier; 2005. p. 209–12.
[15] Valve Sizing Technical Bulletin. Swagelok Company; 2007.
[16] Ki T, Jeong S. Separation of pulse tube losses based on measurement of energy flow and computational fluid dynamics analysis. In: Cryocoolers 17. Los Angeles (California): ICC Press; 2012. p. 161–8.
[17] Radebaugh R. Thermodynamics of regenerative refrigerators. Generation of Low Temperature and It's Applications. Kamakura (Japan): Shonan Technical Center; 2003. p. 1–20.